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UW

Department of Computer Science

Assignment # 02

Course Code: CS-121

Course Title: Digital Logic Design

Total Marks: 20

Discipline/Semester: BSCS- 2^{nd} B

Date/Deadline: 16^{h} April 2025

Assigned Date: 18^{h} April2025.

CLO-1: **Identify and explain** fundamental concepts of digital logic design including basic and universal gates, number systems, binary coded systems, basic concepts of digital system.
CLO-2: **Acquired knowledge** to apply techniques related to the design and analyze of digital electronic circuits including Boolean algebra.

CLO2-GA2-BT-Level C2

Q1: **A**nswers to all questions in detail.

a. Describe Boolean postulates and theorems with relevant examples.
b. Convert the following Boolean expression into:

$$F(A,B,C)=A'BC+AB'C+ABC'$$

i. Canonical Sum of Minterms (SOM)

ii. Canonical Product of Maxterms (POM)

c. Differentiate between Standard SOP and Canonical SOP using truth tables or examples.

d. Demonstrate how to minimize the following functions using Karnaugh Maps

i. $F(A,B,C)=\sum m(0,1,2,5)$ — (3-variable K-map)